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SSAPy - Space Situational Awareness for Python

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Summary

SSAPy is a fast and flexible orbital modeling and analysis tool for orbits spanning from low-Earth into the cislunar regime. Orbits can be flexibly specified from common input format such as Keplerian elements or two-line element data files. SSAPy allows users to model satellites and specify parameters such as satellite area, mass, and drag coefficients. SSAPy includes a customizable force propagation with a range of Earth, Lunar, radiation, atmospheric, and maneuvering models. SSAPy makes use of various community integration methods and can calculate time-evolved orbital quantities, including satellite magnitudes and state vectors. Users can specify various space- and ground-based observation models with support for multiple coordinate and reference frames. SSAPy also supports orbit analysis and propagation methods such as multiple hypothesis tracking and has built-in uncertainty quantification. The majority of SSAPy's methods are vectorized and parallelizable, allowing effective use of high performance computer (HPC) systems. Finally, SSAPy has plotting functionality, allowing users to visualize orbits and trajectories, an example of which is shown in Figure 1.

SSAPy has been used for the classification of cislunar (Higgins et al., 2024), and closely-spaced (Pruett et al., 2024), orbits as well as for studying the long-term stability of orbits in cislunar space (Yeager et al., 2023). SSAPy has also be used to build a case study for rare events analysis in the context of satellites passing close to each other in space (Miller et al., 2022).

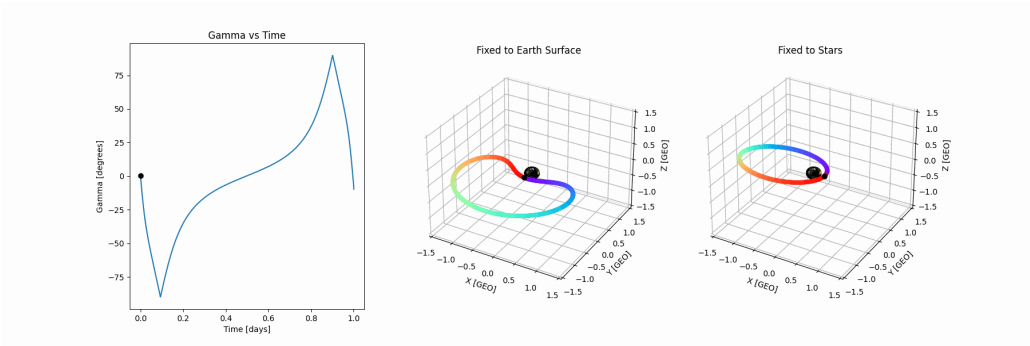

Figure 1: Example SSAPy visualization plots. The left plot shows an example relative angle between an orbit and the Earth's surface and is used for determining reentry points. The middle plot is the corresponding orbit in the International Terrestrial Reference Frame (ITRF), a frame fixed to the Earths surface. The third frame is the orbit in Geocentric Celestial Reference Frame (GCRF), a frame fixed to the stars. Color on the right two plots represents time.

Statement of need

Cislunar space is a region between earth out to beyond the Moon's orbit that includes the Lagrange points. This region of space is of growing importance to space exploration endeavors, scientific (e.g., Duggan et al., 2019) or otherwise. Understanding, mapping, and modeling

orbits through cislunar space is critical to all of these endeavors. The challenge for cislunar orbits is that n-body dynamics (e.g., gravitational forces from the Sun, Earth, Moon and other planets) are significant, leading to unpredictable and chaotic orbital motion. In this chaotic regime, orbits cannot be reduced to simple parametric descriptions making scalable orbit simulation and modeling a critical analysis tool (Yeager et al., 2023). Current orbit modeling software tools are predominantly used via graphical user interfaces (e.g., The General Mission Analysis Tool; Hughes et al. (2014) or the Systems Tool Kit) and are not optimized for large scale simulation on HPC systems. Orbital modeling codes that can be run on HPC systems (e.g., REBOUND; Rein & Liu (2012)) lack full observable generation and modeling capabilities with uncertainty quantification. SAPPy, with its full-featured modeling framework and scalable, parallelizable functionality, fills the gap in the orbital software landscape.

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SSAPy depends on numpy (Harris et al., 2020), scipy (Virtanen et al., 2020), matplotlib (Hunter, 2007), emcee (Foreman-Mackey et al., 2013), astropy (Astropy Collaboration et al., 2022), pyerfa (Kerkwijk et al., 2023), lmfit (Newville et al., 2024), and sqp4 (Vallado et al., 2006). This work was performed under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory (LLNL) under Contract DE-AC52-07NA27344. The document number is LLNL-JRNL-XXXX and the code number is LLNL-CODE-862420. SSAPy was developed with support from LLNL's Laboratory Directed Research and Development Program under projects 19-SI-004 and 22-ERD-054.

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